

PROBLEM SET 3—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: These are my suggested solutions. It is your responsibility to check this document against your own solutions.

1. (a) We can use the *characterization in Proposition 4.14 of FOSD* to answer this question. Note that for any lottery $p = (p_1, p_2, p_3) \in \Delta(Z)$, where Z is as above, we have that the CDF of p is given by

$$F_p(x) = \begin{cases} 0 & x \in (-\infty, 0) \\ p_1 & x \in [0, 1) \\ p_1 + p_2 & x \in [1, 5) \\ 1 & x \in [5, \infty) \end{cases}$$

Thus, if $p = (1/3, 1/3, 1/3)$, $q = (1/4, 1/4, 1/2)$, and $h = (1/5, 3/5, 1/5)$, their CDF's are:

$$F_p(x) = \begin{cases} 0 & x \in (-\infty, 0) \\ 1/3 & x \in [0, 1) \\ 2/3 & x \in [1, 5) \\ 1 & x \in [5, \infty) \end{cases} \quad F_q(x) = \begin{cases} 0 & x \in (-\infty, 0) \\ 1/4 & x \in [0, 1) \\ 1/2 & x \in [1, 5) \\ 1 & x \in [5, \infty) \end{cases} \quad F_h(x) = \begin{cases} 0 & x \in (-\infty, 0) \\ 1/5 & x \in [0, 1) \\ 4/5 & x \in [1, 5) \\ 1 & x \in [5, \infty) \end{cases}$$

From the above you can see that $F_q(x) \leq F_p(x)$ for all x . Thus $q \succ_{\text{fosd}} p$. On the other hand the pairs F_p with F_h , and F_q with F_h cannot be ordered, so p with h and h with q are not ordered by first-order dominance.

- (b) Let $p \in \Delta(Z)$ be an arbitrary lottery and let F_p be the corresponding CDF. A lottery $q = (q_1, q_2, q_3) \in \Delta(Z)$, with cdf F_q , first-order stochastically dominates p if and only if $F_q(x) \leq F_p(x)$ for all x . This implies that the following restrictions on q must hold:

$$\begin{aligned} q_1 &\leq p_1 \\ q_1 + q_2 &\leq p_1 + p_2 \\ q_1 + q_2 + q_3 &= 1 \\ 0 &\leq q_i \leq 1, i = 1, 2, 3. \end{aligned} \tag{1}$$

The first two inequalities come from first-order stochastic dominance and the third equality together with the last trio of inequalities are the restrictions imposed by $q \in \Delta(Z)$. Thus, $\succ_{\text{fosd}}(p) = \{q \in \Delta(Z) | q \text{ satisfies the system (1)}\}$. Since (1) is a system of linear equalities and inequalities which is non-empty, it is *closed and convex*.¹

- (c) The Marshack-Machina triangle, Δ_M is depicted in the figure 1 in green, as the area bounded by the orange lines. The lotteries $p = (1/3, 1/3, 1/3)$, $q = (1/4, 1/4, 1/2)$ and $h = (1/5, 3/5, 1/5)$ are also depicted in the diagram; note that to plot them in the diagram, we get rid of the second coordinate.
- (d) In order to plot $\succ_{\text{fosd}}(p)$ in Δ_M for an arbitrary lottery $p = (p_1, p_2, p_3) \in \Delta(Z)$, we can eliminate p_2 from the linear system (1) to get the equivalent system:

$$\begin{aligned} q_1 &\leq p_1 \\ 1 - q_3 &\leq 1 - p_3 \Leftrightarrow q_3 \geq p_3 \\ 0 &\leq q_1 + q_3 \leq 1 \\ 0 &\leq q_i \leq 1, i = 1, 3. \end{aligned} \tag{2}$$

¹It is a well-known property that any non-empty system of linear equalities and inequalities, otherwise called a *linear system*, defines a closed and convex region. If you are not convinced of this fact, you should be able to provide a simple proof by yourself.

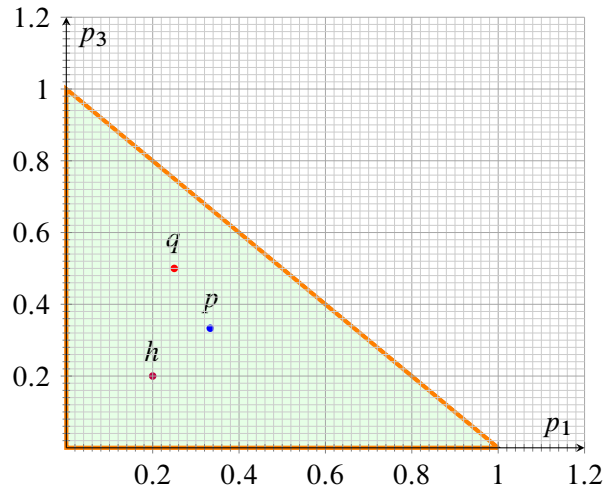


Figure 1: The Marshack-Machina triangle, Δ_M

Thus, the set of lotteries that first-order dominate p are $\succ_{\text{fosd}}(p) = \{q \in \Delta(Z) | q \text{ satisfies (2)}\}$. This region is plotted in figure 2 in red. Note that $\succ_{\text{fosd}}(p)$ is also a closed and convex set.

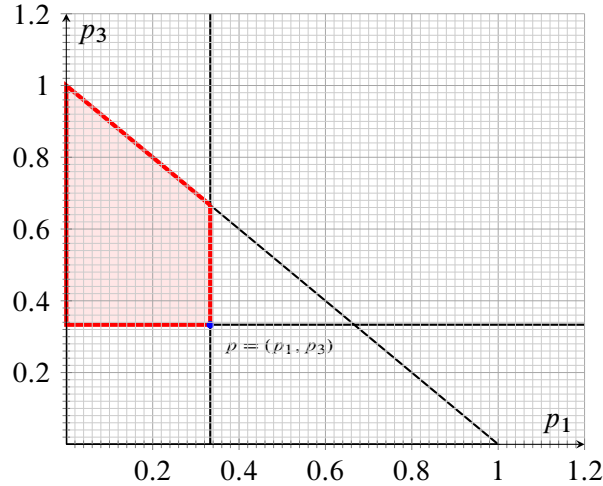


Figure 2: The set $\succ_{\text{fosd}}(p)$ in the Marshack-Machina triangle Δ_M

- (e) Suppose that a lottery $p = (p_1, p_2, p_3)$ has expected value equal to E_p . Then, the lottery $q = (q_1, q_2, q_3)$ has an expected value equal to E_p when $q_1(0) + q_2(1) + q_3(5) = q_2 + 5q_3 = E_p$. Since $q_2 = 1 - q_1 - q_3$, we can eliminate it from the previous equation to get $1 - q_1 + 4q_3 = E_p$. Solving for q_3 we get the equation of the iso-expected value line through p :

$$q_3 = \frac{E_p - 1}{4} + \frac{q_1}{4}. \quad (3)$$

Note that the slope of the iso-expected value line (3) is equal to $\frac{1-0}{5-1} = \frac{1}{4}$, where the numerator is the difference between the second and the first prize and the denominator is the difference between the third and the second prize. Figure 3 plots the iso-expected value lines for $p = (1/3, 1/3, 1/3)$, $q = (1/4, 1/4, 1/2)$ and $h = (1/5, 3/5, 1/5)$.

- (f) In lecture 4, definition 4.16 gave a definition of second order stochastic dominance. Later, in proposition 4.18, second order stochastic dominance was characterized in terms of the equivalence of three properties for lotteries with *the same means*. However, the result in proposition 4.18 can be extended to lotteries with *different means* to give a *complete characterization* of SOSD. Theorem PS3.1 below states this result. The notation is the same as in the lecture.

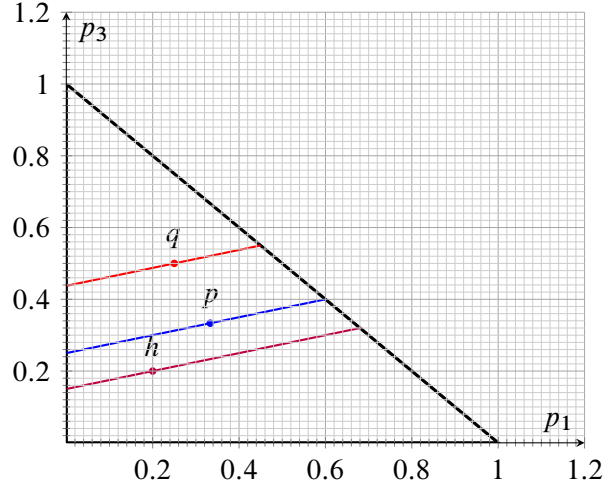


Figure 3: Iso-expected value lines for the lotteries p , q and h given at the start of the problem.

Theorem PS3.1. Let $Z = [w, b]$ be an infinite set of prizes and let $\mathfrak{F}$ be the set of CDFs (lotteries) over Z . Let $F, G \in \mathfrak{F}$. Then, the following are equivalent.

- i. F second-order stochastically dominates G written as $F \succ_{\text{sosd}} G$.
- ii. For every non-decreasing concave function $u : Z \rightarrow \mathbb{R}$, we have $\int u(z) dF(Z) \geq \int u(z) dG(z)$.
- iii. There exist random variables X and Y such that X has CDF F and Y has CDF G , and such that $Y = X + \epsilon$, where ϵ is a random variable that has conditional mean less than or equal to zero, conditional on X , that is, $\mathbb{E}[\epsilon|X] \leq 0$. In words, X is equal to Y plus a supplement which can be negative or positive but that, conditional on X , must have non-positive mean.
- iv. For every $t \geq 0$, $\int_w^t G(x) dx \geq \int_w^t F(x) dx$.

The definition in (iii) in theorem PS3.1 above generalizes the notion of a mean-preserving spread definition given in definition 4.17 of lecture 4. It is useful to compare Theorem PS3.1 with proposition 4.18 from lecture 4. Notice that everything remains the same except for item (iii) in the theorem, in which the notion of a mean-preserving spread is extended to handle lotteries with different means. In particular, note that the integral of the CDF characterization of second order stochastic dominance remains the same.

To characterize the set $\succ_{\text{sosd}}(p)$ for any arbitrary lottery $p \in \Delta(Z)$, we are going to use the integral of the CDF characterization from theorem PS3.1 (iv) above. To this end, let $f = (f_1, f_2, f_3)$ be a lottery in $\Delta(M)$ with corresponding CDF F and similarly for $g = (g_1, g_2, g_3)$ and G . Recall, from problem set 2 solutions, that the integral of the CDFs of F will always take the following shape:

$$\begin{aligned} \zeta_F(x) &= \int_{-\infty}^x F(s) ds \\ &= \begin{cases} 0 & x \in (-\infty, 0] \\ f_1 x & x \in (0, 1] \\ f_1 + (f_1 + f_2)(x - 1) & x \in (1, 5] \\ x - \mathbb{E}[F] & x \in (5, \infty) \end{cases} \\ &= \begin{cases} 0 & x \in (-\infty, 0] \\ f_1 x & x \in (0, 1] \\ f_1 + (1 - f_3)(x - 1) & x \in (1, 5] \\ x - \mathbb{E}[F] & x \in (5, \infty) \end{cases} \end{aligned}$$

where the third line comes from the fact that $f_1 + f_2 = 1 - f_3$, so that we can eventually graph

our results in the Marshack-Machina triangle. Similarly for G we obtain:

$$\begin{aligned}\zeta_G(x) &= \int_{-\infty}^x F(s)ds \\ &= \begin{cases} 0 & x \in (-\infty, 0] \\ g_1 x & x \in (0, 1] \\ g_1 + (g_1 + g_2)(x - 1) & x \in (1, 5] \\ x - \mathbf{E}[G] & x \in (5, \infty) \end{cases} \\ &= \begin{cases} 0 & x \in (-\infty, 0] \\ g_1 x & x \in (0, 1] \\ g_1 + (1 - g_3)(x - 1) & x \in (1, 5] \\ x - \mathbf{E}[G] & x \in (5, \infty) \end{cases}\end{aligned}$$

Note that the integral of the CDF functions are always continuous² and it is also a convex function.

Now, we have that $F \succ_{\text{sosd}} G$ if and only if $\zeta_F(x) \leq \zeta_G(x)$ for all x . This implies that

$$f_1 x \leq g_1 x \quad \forall x \in (0, 1] \Leftrightarrow f_1 \leq g_1 \quad (4)$$

$$f_1 + (1 - f_3)(x - 1) \leq g_1 + (1 - g_3)(x - 1) \quad \forall x \in (1, 5] \quad (5)$$

$$x - \mathbf{E}[F] \leq x - \mathbf{E}[G] \quad \forall x \in (5, \infty) \Leftrightarrow \mathbf{E}[G] \leq \mathbf{E}[F]. \quad (6)$$

Note that (4) was simplified to the inequality in the right hand side because the left hand side was valid for all $x \in (0, 1]$. Similarly with the inequality in (6). However, the inequality in (5) is a collection of an *infinite amount* of inequalities, one for each $x \in (1, 5]$. However, I will now show that (5) is a *redundant* inequality if (4) and (6) hold. We need some preparations. Note that:

$$\mathbf{E}[G] = g_1(0) + g_2(1) + g_3(5) = g_2 + 5g_3 = (1 - g_1 - g_3) + 5g_3 = 1 - g_1 + 4g_3,$$

$$\mathbf{E}[F] = f_1(0) + f_2(1) + f_3(5) = (1 - f_1 - f_3) + 5f_3 = 1 - f_1 + 4f_3,$$

and using the above, we can rewrite (6), as follows:

$$\mathbf{E}[G] \leq \mathbf{E}[F] \Leftrightarrow 1 - g_1 + 4g_3 \leq 1 - f_1 + 4f_3 \Leftrightarrow 4(g_3 - f_3) \leq g_1 - f_1 \quad (7)$$

Now, (4) implies that $g_1 - f_1 \geq 0$, therefore, (4) together with (7) is equivalent to:

$$\frac{g_3 - f_3}{g_1 - f_1} \leq \frac{1}{4} \quad (8)$$

With the above preparations, we are ready to show that (4) and (7) imply (5), i.e., (5) is redundant. Note that, using (4), we can rewrite (5) as:

$$\frac{g_3 - f_3}{g_1 - f_1} \leq \frac{1}{x - 1} \quad \forall x \in (1, 5], \quad (9)$$

but note that (9) is equivalent to

$$\frac{g_3 - f_3}{g_1 - f_1} \leq \min \left\{ \frac{1}{x - 1} \mid x \in (1, 5] \right\} = 1/4. \quad (10)$$

Note that (10) is equivalent to (9) because if the inequality holds for the minimal value of the right-hand side over $x \in (1, 5]$, then it holds for all x over the same interval. But (10) is equivalent to (4) and (7). Thus, (5) is redundant.

²Because of the continuity of the integral of the CDF function it doesn't matter how we write the limits in its piecewise definition. That is, in the definitions above, we could have used left-closed and right-open intervals (like I wrote them in the PS2 solutions), instead of left-open and right-closed as it is written above.

Let $p = (p_1, p_2, p_3) \in \Delta(Z)$ be any arbitrary lottery. Joining (4) and (5) with the inequalities describing the set of lotteries in the Marshack-Machina Triangle, we get the following system of equations that characterizes $\succsim_{\text{sosd}}(p)$, i.e., $q = (q_1, q_2, q_3) \in \succsim_{\text{sosd}}(p)$ if q satisfies:

$$\begin{aligned} p_1 &\leq q_1 \\ \frac{p_3 - q_3}{p_1 - q_1} &\leq \frac{1}{4} \\ 0 &\leq q_1 + q_3 \leq 1 \\ 0 &\leq q_i \leq 1, i = 1, 3. \end{aligned} \tag{11}$$

Once again, since the linear system (11) characterizing $\succsim_{\text{sosd}}(p)$ is non-empty, then it is closed and convex. In the Marshack-Machina triangle, $\succsim_{\text{sosd}}(p)$ is represented in figure 4 with the orange area. In the case when the lotteries have *equal means*, $\succsim_{\text{sosd}} p$ corresponds only to the blue line in 4.

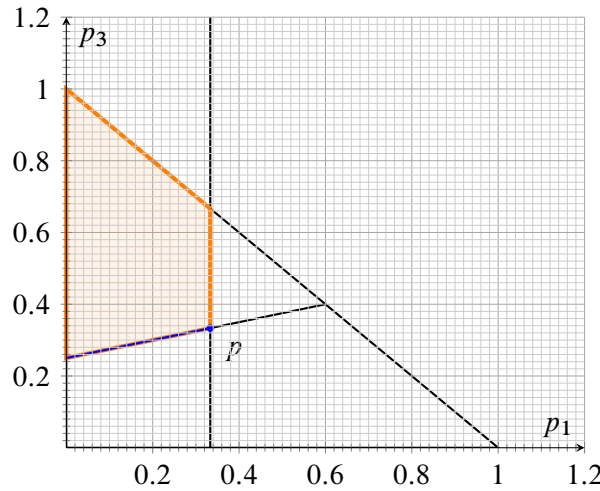


Figure 4: The orange area is the set of lotteries that SOSD p . The blue line is the case when we are only considering lotteries with equal means to p .

Remark: If you look carefully at figures 2 and 4 you will see that $\succsim_{\text{fosd}}(p) \subset \succsim_{\text{sosd}}(p)$. This is a general result, and it means that FOSD implies SOSD, but not vice versa.

- (g) Let $u : Z \rightarrow \mathbb{R}$ be an increasing Bernoulli utility function with respect to money. Let $p \in \Delta(Z)$ be an arbitrary lottery over Z . Note that since u is increasing, we have that $u(0) < u(1) < u(5)$. The following equation, defines *implicitly* an indifference curve through p :

$$p_1 u(0) + p_2 u(1) + p_3 u(5) = E[u(p)]$$

Rewriting the above with $p_2 = 1 - p_1 - p_3$ and solving for p_3 yields the equation of the indifference curve that can be plotted in the Marshack-Machina triangle:

$$p_1 u(0) + (1 - p_1 - p_3) u(1) + p_3 u(5) = E[u(p)] \Rightarrow p_3 = \frac{E[u(p)] - u(1)}{u(5) - u(1)} + \frac{u(1) - u(0)}{u(5) - u(1)} p_1 \tag{12}$$

Note that the indifference curve through p is an *affine* function, i.e., a linear part plus a constant. The slope $\frac{u(1) - u(0)}{u(5) - u(1)}$ is strictly positive because $u(0) < u(1) < u(5)$. Figure 5 plots an indifference curve through $p = (1/3, 1/3, 1/3)$ for some Bernoulli utility u .

- (h) We can say a lot more about the slope of the indifference curve through p , from (12), using the *Three Strings Lemma* from page 12 in lecture 4:

u exhibits risk aversion In this case u is concave so the three strings lemma implies that for $0 < 1 < 5$

$$\frac{u(1) - u(0)}{u(5) - u(1)} \geq \frac{1 - 0}{5 - 1} = \frac{1}{4}.$$

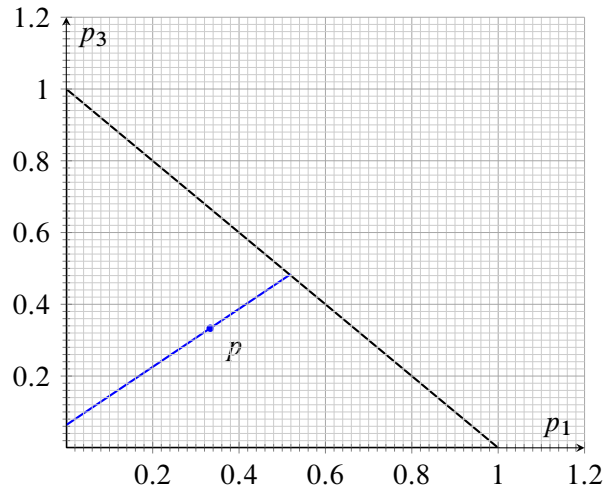


Figure 5: Indifference curve through $p = (1/3, 1/3, 1/3)$.

The above implies that if the agent is risk averse and thus has a concave u , the slope of the indifference curve through p is *steeper* than the iso-expected value line.

u exhibits risk loving In this case, u is convex so the inequality in the three strings lemma is reversed and thus we have that:

$$\frac{u(1) - u(0)}{u(5) - u(1)} \leq \frac{1 - 0}{5 - 1} = \frac{1}{4}.$$

Thus, a risk loving individual who has a convex u , has indifference curves that are *less steep* than the iso-expected value line.

u exhibits risk neutrality Risk neutrality combines the two cases above to form an equality:

$$\frac{u(1) - u(0)}{u(5) - u(1)} = \frac{1 - 0}{5 - 1} = \frac{1}{4}.$$

Thus, a risk neutral agent has indifference curves with a slope *equal* to the iso-expected value line.

In figure 6 I plot the indifference curve for three different decision makers who are risk-averse, risk-loving, and risk neutral.

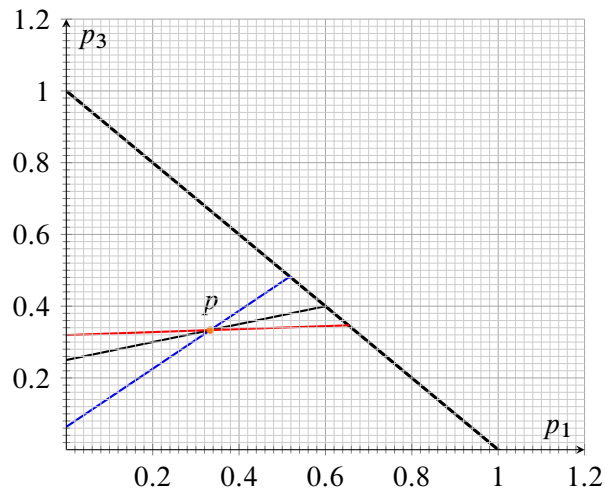


Figure 6: Indifference curves through $p = (1/3, 1/3, 1/3)$ for different decision makers: the **blue line** represents a risk-averse DM, the **red line** a risk-loving DM, and the black line a risk-neutral DM.

- (i) If u_1 is more risk averse than u_2 , then proposition 4.12 from lecture 4 will help us characterize the indifference curves for these two decision makers. The fact that u_1 is more risk averse than u_2 implies that there exists an increasing and concave function Ψ such that $u_1(x) = \Psi(u_2(x))$ for all x . Since u is increasing, as well as Ψ , we have the following inequalities:

$$u_2(0) < u_2(1) < u_2(5) \\ \Psi(u_2(0)) < \Psi(u_2(1)) < \Psi(u_2(5))$$

The concavity of u , Ψ and the inequalities above imply that we can apply the three strings lemma twice to get:

$$\frac{u_1(1) - u_1(0)}{u_1(5) - u_1(1)} = \frac{\Psi(u_2(1)) - \Psi(u_2(0))}{\Psi(u_2(5)) - \Psi(u_2(1))} > \frac{u_2(1) - u_2(0)}{u_2(5) - u_2(1)} > \frac{1 - 0}{5 - 1} = \frac{1}{4}$$

Recall from equation (12) for an arbitrary Bernoulli utility function u , that the slope is precisely given by the quotients above. Therefore, we can conclude that if u_1 is more risk-averse than u_2 , the indifference curve through lottery p for u_1 is steeper than the one for u_2 . Figure 7 plots the indifference curves through $p = (1/3, 1/3, 1/3)$ for $u_1(z) = z^{1/3}$ and for $u_2 = z^{1/2}$. Note that the concave transformation that converts u_2 to u_1 is $\Psi(x) = x^{2/3}$.

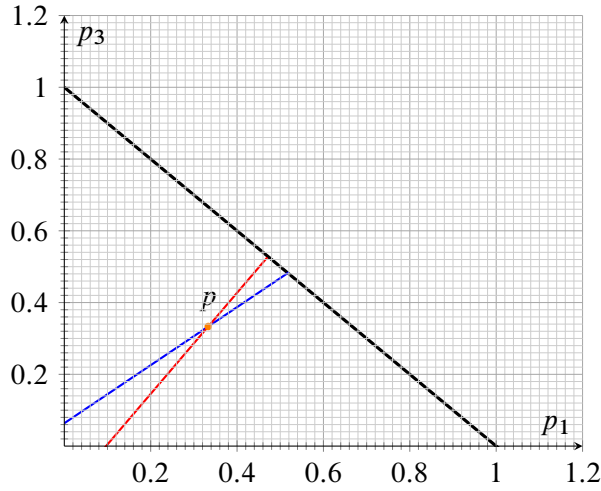


Figure 7: Indifference curves through $p = (1/3, 1/3, 1/3)$ for two decision makers: the red IC with u_1 is more risk averse than the blue IC with u_2 .

2. The stocks α and β can be seen as *finitely supported lotteries over the real line*. With a slight abuse of notation, let us represent their probability mass functions by $\alpha = (7, 12, 14; 0.3, 0.4, 0.3)$ and $\beta = (15, 5, 10; 0.3, 0.4, 0.3)$, where the first three elements denote the monetary return on the stock and the next three elements correspond to the probabilities with which each return obtains.

- (a) Let F denote the CDF of α and G denote the CDF of β . We have that:

$$F(x) = \begin{cases} 0 & x < 7 \\ 0.3 & 7 \leq x < 12 \\ 0.7 & 12 \leq x < 14 \\ 1 & x \geq 14 \end{cases} \quad G(x) = \begin{cases} 0 & x < 5 \\ 0.3 & 5 \leq x < 10 \\ 0.7 & 10 \leq x < 15 \\ 1 & x \geq 15 \end{cases}$$

The graphs of the CDFs are in figure 8 below. Note that it is not the case that $F(x) \leq G(x)$ for all x nor $G(x) \geq F(x)$ for all x . Thus, the lotteries cannot be ordered according to first order stochastic dominance.

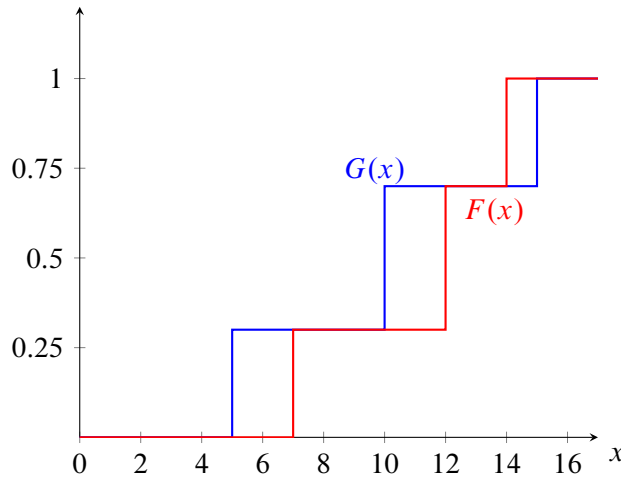


Figure 8: CDFs F and G of α and β respectively.

- (b) We can use the integral of the CDF characterization to answer this question. Similarly as we did in the last question, these functions are:

$$\zeta_F(x) = \begin{cases} 0 & x \leq 7 \\ 0.3x - 2.1 & 7 < x \leq 12 \\ 0.7x - 6.9 & 12 < x \leq 14 \\ x - 11.1 & x > 14 \end{cases} \quad \zeta_G(x) = \begin{cases} 0 & x \leq 5 \\ 0.4x - 2 & 5 < x \leq 10 \\ 0.7x - 5 & 10 < x \leq 15 \\ x - 9.5 & x > 15 \end{cases}$$

Both functions are plotted in figure 9 below. Note that $\zeta_G(x) \geq \zeta_F(x)$ for all x . Therefore, according to the characterization in theorem PS3.1, we have that $\alpha \succ_{\text{sosd}} \beta$. Note the power of theorem PS3.1: although α and β do not have the same mean, nevertheless we are able to order them according to second order dominance.

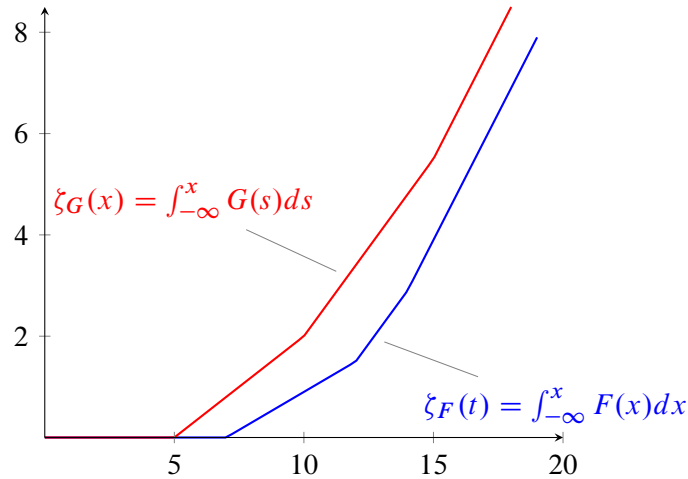


Figure 9: Integral of CDF functions

There is another way to check SOSD which is using the decomposition in part (iii) of theorem PS3.1. That is, if Y is the random variable representing β and X is the random variable representing α , we must show that we can write

$$Y = X + \epsilon$$

where ϵ is a random variable that can be positive or negative, but conditional on X , it must satisfy $E[\epsilon|X] \leq 0$. Once again, note that if the lotteries have the same mean we, then ϵ must satisfy $E[\epsilon|X] = 0$, i.e., a mean-preserving spread, according to proposition 4.18, part (ii) in lecture 4.

However, since α and β do not have the same mean, we do need the result of theorem PS3.1. Note that X , being the random variable representing α , can take on the values $\{7, 12, 14\}$. Suppose that ϵ is a random variable that can take the values $\{-4, -2, 1, 3\}$; the reason for taking this values will become clear in a moment. Below, in table 1 I describe the conditional distribution of ϵ given X : The numbers $\{-4, -2, 1, 3\}$ so as to *split* the original monetary prizes of X : we are

ϵX	$X = 7$	$X = 12$	$X = 14$
-4	0	0	θ
-2	η	λ	0
1	0	0	$1 - \theta$
3	$1 - \eta$	$1 - \lambda$	0

Table 1: The conditional distribution of $\epsilon|X$

constructing a new lottery in two steps, first the outcome of X is realized, and then given the value of X , we pick a value of $\epsilon|X$ according to the distribution in table 1 that is added up as an extra. What this two step lottery is doing is *splitting and constructing* the monetary prizes of Y : If $X = 7$, then the second either subtracts -2 to yield a final outcome of 5 , or adds 3 to yield a final outcome of 10 . If $X = 12$, then the second stage lottery subtracts -2 to yield 10 or adds 3 to yield 15 . Finally, if $X = 14$, the second stage subtracts -4 to get 10 or adds 1 to get 15 .

The unconditional distribution of ϵ is obtained using the law of total probability:

$$\Pr(\epsilon) = \sum_x \Pr(\epsilon|X = x) \Pr(X = x),$$

this distribution is given in table 2. The triple of numbers (η, λ, θ) are chosen to satisfy the

ϵ	$\Pr(\epsilon)$
-4	0.3θ
-2	$0.3\eta + 0.4\lambda$
1	$0.3(1 - \theta)$
3	$0.3(1 - \eta) + 0.4(1 - \lambda)$

Table 2: Unconditional distribution of ϵ .

following inequalities:

$$\mathbb{E}[\epsilon|X = 7] = \eta(-2) + (1 - \eta)(3) \leq 0 \quad (13)$$

$$\mathbb{E}[\epsilon|X = 12] = \lambda(-2) + (1 - \lambda)(3) \leq 0 \quad (14)$$

$$\mathbb{E}[\epsilon|X = 14] = \theta(-4) + (1 - \theta)(1) \leq 0. \quad (15)$$

Also, since $\mathbb{E}[Y] = \mathbb{E}[X] + \mathbb{E}[\epsilon]$ and $\mathbb{E}[Y] = 9.5$ and $\mathbb{E}[X] = 11.1$, then we must also have

$$\mathbb{E}[\epsilon] = 0.3\theta(-4) + (0.3\eta + 0.4\lambda)(-2) + (0.3(1 - \theta))(1) + (0.3(1 - \eta) + 0.4(1 - \lambda))(3) = -1.6. \quad (16)$$

Any triple (η, λ, θ) satisfying (13), (14), (15), and (16), will yield the required random variable ϵ . For example, the numbers $\eta = 0.6$, $\lambda = 0.95$, and $\theta = 0.8$, satisfy the required conditions and constitute the required random variable ϵ , as you can readily check.

- (c) Again, with a slight abuse of notation, let α denote the return on the stock of the same name. Since the investor has already satisfied all required consumption and savings yield no interest, two things can happen: either the investor waits and gets the random return $w + \alpha$ on period t_1 , or she sells the stock in period t_0 to get the sure return $w + P$, where P is the prize at which the investor sells the stock. Then, the minimum price $P(w)$ at which the investor is willing to sell is given by solving the following equation:

$$u(w + P(w)) = \mathbb{E}[u(w + \alpha)] \Leftrightarrow \sqrt{w - P(w)} = 0.3\sqrt{w + 7} + 0.4\sqrt{w + 12} + 0.3\sqrt{w + 14} \quad (17)$$

i. If $w = 10$, equation (17) becomes:

$$\sqrt{10 + P(w)} = 4.58279 \Leftrightarrow P(w) = 11.002$$

ii. If $w = 10000$, equation (17) becomes:

$$\sqrt{10000 + P(w)} = 100.055 \Leftrightarrow P(w) = 11.0998$$

What is going on in here? We can calculate the *risk premium* $RF(w) = E[\alpha] - P(w) = 11.1 - P(w)$. Since we have that

$$P(w) = \left(0.3\sqrt{w+7} + 0.4\sqrt{w+12} + 0.3\sqrt{w+14}\right)^2 - w,$$

then we have that

$$RF(w) = 11.1 - \left(0.3\sqrt{w+7} + 0.4\sqrt{w+12} + 0.3\sqrt{w+14}\right)^2 + w.$$

A plot of it is in figure 10. Recall that the risk premium is the *price that the DM would be willing to pay to get rid of risk*. Notice that it is decreasing in w . This means that $P(w)$ approaches the expected value of the return of the stock, 11.1, as w grows large. Why the investor is willing to do this as wealth increases? The answer can be found by calculating the coefficient of absolute risk aversion for $u(w) = \sqrt{w}$. This coefficient is:

$$r_A(w) = -\frac{-(1/4)w^{-3/2}}{(1/2)w^{-1/2}} = \frac{1}{2w}$$

Notice that $r_A(w)$ is decreasing in wealth. This means that $u(w) = \sqrt{w}$ exhibits *Decreasing Absolute Risk Aversion* with respect to wealth. Intuitively, the decision maker becomes “less risk averse” as the wealth increases. If you plot $u(w) = \sqrt{w}$ you will notice that its curvature decreases with w . That is why the investor in the problem has a decreasing risk premium with respect to wealth, or the price that he is willing to accept to part with the stock is increasing but at a decreasing rate and it approaches the expected value of 11.1 of the stock α .

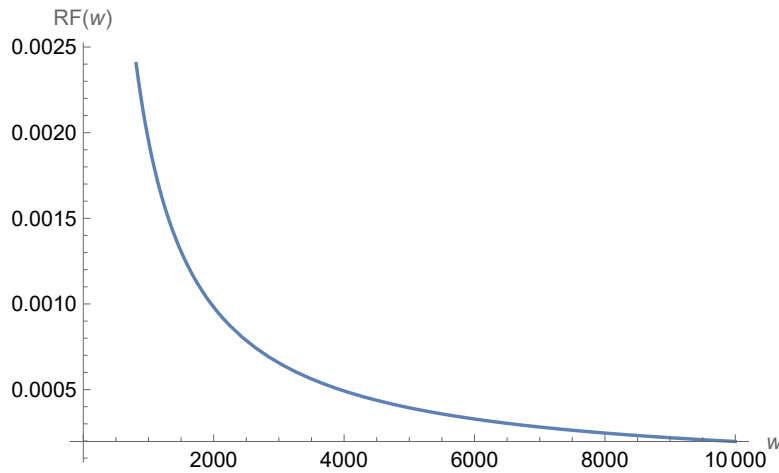


Figure 10: The risk premium $RF(w)$ as a function of wealth.